

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_192147_g
- started_at_local: 2026-05-13 19:21:47
- finished_at_local: 2026-05-13 19:24:46
- duration_hhmm: 00:03
- duration_seconds: 179.394
- seed_offset: 6000
- replicate_label: g
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.22093	0.05147	0.145614	0.0	0.78	0.0
cvrp_random_replay	random	score_only	False	0.241884	0.005376	0.136769	0.0	0.58	0.0
cvrp_failure_replay	failure	score_only	False	0.232866	0.007671	0.132779	0.874439	0.78	0.005256
cvrp_failure_replay_compression	failure	novelty_gate	True	0.245806	0.005376	0.138948	0.0	0.58	0.0

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.22093, synthetic 0.05147, combined 0.145614.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.22093 across 5 instances; family means: A=0.206806, B=0.194209, E=0.21881, P=0.290614.
- Panel synthetic_holdout mean gap 0.05147 across 4 instances; family means: alternating_belt=0.049043, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.156836.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 544, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.187773}, {"best_known_cost": 937, "cost": 1083, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.155816}, {"best_known_cost": 784, "cost": 882, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.125}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.241884, synthetic 0.005376, combined 0.136769.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.241884 across 5 instances; family means: A=0.143979, B=0.254873, E=0.238004, P=0.31769.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 587, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.281659}, {"best_known_cost": 937, "cost": 1062, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.133404}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.232866, synthetic 0.007671, combined 0.132779.
- Accepted-epoch count 2, mean accepted code novelty 0.874439, and final complexity 0.78.
- Adaptation efficiency 0.005256 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.232866 across 5 instances; family means: A=0.122164, B=0.19963, E=0.247601, P=0.395307.
- Panel synthetic_holdout mean gap 0.007671 across 4 instances; family means: alternating_belt=0.030685, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 613, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.338428}, {"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.245806, synthetic 0.005376, combined 0.138948.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.245806 across 5 instances; family means: A=0.143979, B=0.248809, E=0.285988, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.145144}, {"best_known_cost": 784, "cost": 858, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.094388}]

Judge Appendix

Summary of Replay-Aware CVRP Benchmark Suite Results

Condition	Replay Mode	Heldout CVRPLIB Gap	Synthetic Holdout Gap	Transfer Gap	Code Novelty Mean	Mean Complexity	Comments on Transfer & Novelty
cvrp_no_replay	None	**0.22093**	0.05147	0.14561	0.0	0.78	Best transfer gap among conditions with zero code novelty.
cvrp_random_replay	Random replay	0.24188	**0.00538**	0.13677	0.0	0.58	Best synthetic holdout gap but higher heldout CVRPLIB gap; no code novelty.
cvrp_failure_replay	Failure replay	0.23287	0.00767	**0.13278**	**0.8744**	0.78	Best transfer gap and highest code novelty; however, heldout gaps higher than no replay.
cvrp_failure_replay_compression	Failure replay + compression	0.24581	0.00538	0.13895	0.0	0.58	Compression lowers complexity; transfer gap worse than failure replay without compression and no code novelty.

Interpretation:

- 1. **Transfer Evidence (Heldout CVRPLIB & Synthetic Gaps)****
 - ****Best heldout CVRPLIB gap****: ***cvrp_no_replay*** (0.22093), indicating strongest direct transfer to heldout CVRPLIB instances.
 - ****Best synthetic holdout gap****: ***cvrp_random_replay*** and ***cvrp_failure_replay_compression*** (0.0054), suggesting best adaptation to synthetic problem variations.
 - ****Best combined transfer gap****: ***cvrp_failure_replay*** (0.13278), slightly outperforming ***cvrp_no_replay*** (0.14561) and ***cvrp_random_replay*** (0.13677).
- 2. **Replay Mode Effects****
 - ***No replay*** yields the strongest CVRPLIB heldout performance and balanced transfer gap.
 - ***Random replay*** significantly improves synthetic transfer (gap 0.0054) but worsens CVRPLIB heldout gap (0.2419).

- *Failure replay* improves transfer gap slightly over no replay but results in poorer heldout CVRPLIB (0.2329 vs 0.2209).
 - Adding *compression* to failure replay reduces complexity but degrades transfer performance and eliminates code novelty.
3. ****Code Novelty vs Transfer:****
- *Failure replay* results in high code novelty (0.8744 mean) but transfer improvements are modest and heldout CVRPLIB gap is not better than no replay.
 - Other conditions have zero code novelty; thus, lexical or algorithmic novelty claims are unsupported except for failure replay.
 - However, improved transfer metrics under failure replay occur despite increased code novelty, contrasting with *no replay* and *random replay*.
4. ****Complexity:****
- Replay modes introducing compression lowered complexity (0.58), but this did not translate into improved transfer or heldout CVRPLIB gaps.
 - Balanced profiles had higher complexity (0.78).

Conservative Conclusion:

- The ****no replay**** condition provides the best ****heldout CVRPLIB optimality gap****, thus strongest conservative evidence for transfer to standard benchmark instances.
- The ****random replay**** condition improves synthetic holdout performance dramatically (lowest synthetic gap), suggesting better adaptation to synthetic distributions but at the cost of worsening heldout CVRPLIB gaps.
- The ****failure replay**** condition achieves the best combined ****transfer gap**** slightly and shows high code novelty, indicating some algorithmic innovation; however, this does not yet translate into robust improvements on the established CVRPLIB heldout benchmark.
- Compression-aware replay lowers complexity but does not improve transfer performance nor code novelty.
- Code novelty and algorithmic improvements in failure replay do not clearly correspond to improved heldout CVRPLIB gaps; thus, lexical novelty does not equate to algorithmic invention in aggregate transfer.

Final prioritized ranking:

Metric	Best Condition	Notes
Heldout CVRPLIB gap	**cvrp_no_replay**	Strongest transfer on realistic benchmarks.
Synthetic holdout gap	**cvrp_random_replay**	Best synthetic adaptation but weaker heldout.
Transfer gap (combined)	**cvrp_failure_replay**	Slightly better transfer gap; highest novelty but heldout gap worse than no replay.

Recommendation for benchmark suite:

- Prioritize ****no replay**** and ****failure replay**** modes when targeting overall transfer, with caution interpreting failure replay novelty due to unclear heldout benefit.
- Random replay benefits synthetic domains but may harm real-world benchmark transfer.
- Compression-aware replay not justified based on current transfer metrics.
- Do not claim algorithmic novelty without consistent improvement in heldout CVRPLIB gaps despite high code novelty under failure replay.